

STIX Version 2.1 Errata 01

Committee Specification Draft 01

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Related work:

This document describes the changes made to:

- *STIX Version 2.1*. Edited by Bret Jordan, Rich Piazza, and Trey Darley. Latest stage: <https://docs.oasis-open.org/cti/stix/v2.1/stix-v2.1.html> (<https://docs.oasis-open.org/cti/stix/v2.1/stix-v2.1.html>).

This specification is related to:

- *TAXII Version 2.1*. Edited by Bret Jordan and Drew Varner. Latest stage: <https://docs.oasis-open.org/cti/taxii/v2.1/taxii-v2.1.html> (<https://docs.oasis-open.org/cti/taxii/v2.1/taxii-v2.1.html>).
- *STIX/TAXII 2.0 Interoperability Test Document: Part 1 Version 1.1*. Edited by Allan Thomson and Jason Keirstead. Latest stage: <https://docs.oasis-open.org/cti/stix-taxii-2-interop-p1/v1.1/stix-taxii-2-interop-p1-v1.1.html> (<https://docs.oasis-open.org/cti/stix-taxii-2-interop-p1/v1.1/stix-taxii-2-interop-p1-v1.1.html>).
- *STIX/TAXII 2.0 Interoperability Test Document: Part 2 Version 1.0*. Edited by Allan Thomson and Jason Keirstead. Latest stage: <https://docs.oasis-open.org/cti/stix-taxii-2-interop-p2/v1.0/stix-taxii-2-interop-p2-v1.0.html> (<https://docs.oasis-open.org/cti/stix-taxii-2-interop-p2/v1.0/stix-taxii-2-interop-p2-v1.0.html>).

Abstract:

Structured Threat Information Expression (STIX) is a language for expressing cyber threat and observable information. This document defines concepts that apply across all of STIX and defines the overall structure of the STIX language.

Status:

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Citation format:

When referencing this specification the following citation format should be used:

[STI-v2.1-errata01]

STIX Version 2.1 Errata 01. Edited by Rich Piazza, Emily Ratliff, Stephan Relitz and Christian Studer. 02 April 2025. OASIS Committee Specification Draft 01. <https://docs.oasis-open.org/cti/stix/v2.1/errata01/csd01/stix-v2.1-errata01-csd01-complete.html> (<https://docs.oasis-open.org/cti/stix/v2.1/errata01/csd01/stix-v2.1-errata01-csd01-complete.html>). Latest stage: <https://docs.oasis-open.org/cti/stix/v2.1/stix-v2.1.html> (<https://docs.oasis-open.org/cti/stix/v2.1/stix-v2.1.html>).

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1. Introduction

This document lists all the corrections made to STIX Version 2.1.

1.1 Scope of changes

Non-material corrections have been made to the STIX 2.1 specification to address issues identified or reported by participants to the TC, listed in the Github issues system, and discussed during CTI TC working call sessions. Changes provide additional or missing vocabulary values, fix typos, improve descriptions, correct examples or provide links between related sections to enhance navigation through the document.

1.2 Description of changes

STIX 2.1 Errata 01 differs from STIX 2.1 in the following ways:

- Updated Malware Embedded Relationships table with missing property in *section 4.11.2 - Malware Relationships*.
 - **operating_system_refs** is no longer missing in the table.
- Malware Analysis Relationships table fixed with the right relationship type between Malware Analysis and Malware in *section 4.12.2 - Malware Analysis Relationships*.
 - wrong **analysis-of** is replaced with the right **av-analysis-of** relationship type.
- Property descriptions fixed in *section 6.6.1 - Email Message Object Properties*.
 - **from_ref**, **sender_ref**, **to_refs**, **cc_refs** and **bcc_refs** properties descriptions now mention the right **email-addr** type they are referencing.
- Examples fixed in *section 6.12.2.1 - HTTP Request Extension Properties*.
 - the **request_header** property is now a **list** of type **string** in the examples, as expected from the description.
- Example fixed in *section 9.5.1 - Observation Expression Qualifiers*.

- the example used to illustrate the use of *Observation Expression* **WITHIN** x **SECONDS** now has the right **windows-registry-key** Observable type.
- Updated *section 10.9 - Implementation Language Vocabulary*.
 - rust** value was added.
- Updated *section 10.11 - Industry Sector Vocabulary*.
 - legal** value was added.
- Fixed summary in *section 10.12 - Infrastructure Type Vocabulary*.
 - missing **control-system**, **firewall**, **routers-switches** and **workstation** values were added to the Summary as they were already described in the Vocabulary table.
- Enhanced descriptions in *section 10.14 - Malware Result Vocabulary*.
 - descriptions for every vocabulary values were improved with more descriptive definitions.
- Fixed missing value in *section 10.22 - Report Type Vocabulary*.
 - incident** value was added.
- Updated *section 10.23 - Threat Actor Type Vocabulary*.
 - private-sector** value was added.
- Fixed multiple Enumeration headers
 - Enumerations now have the right headers, to differentiate enumerations from vocabularies, including:
 - Enumeration Name** is now used instead of **Vocabulary Name**
 - Enumeration Summary** is now used instead of **Vocabulary Summary**
 - Enumeration Value** is now used instead of **Vocabulary Value**
 - These changes apply on:
 - section 10.4 - Encryption Algorithm Enumeration*
 - section 10.5 - Extension Type Enumeration*
 - section 10.16 - Network Socket Address Family Enumeration*
 - section 10.17 - Network Socket Type Enumeration*
 - section 10.18 - Opinion Enumeration*
 - section 10.27 - Windows™ Integrity Level Enumeration*
 - section 10.29 - Windows™ Registry Datatype Enumeration*
 - section 10.30 - Windows™ Service Start Type Enumeration*
 - section 10.31 - Windows™ Service Type Enumeration*
 - section 10.32 - Windows™ Service Status Enumeration*
- Updated *Appendix B: Relationship Summary Table*.
 - Duplicated relationship **located-at** between **threat-actor** and **location** has been removed.
 - Misspelled relationship **exfiltrates-to** between **malware** and **infrastructure** has been fixed.
 - Missing relationships have been added, including:
 - remediates** between **course-of-action** and **malware**
 - remediates** between **course-of-action** and **vulnerability**
 - uses** between **tool** and **infrastructure**
 - resolves-to** between **domain-name** and **domain-name**
 - resolves-to** between **domain-name** and **ipv4-addr**
 - resolves-to** between **domain-name** and **ipv6-addr**
 - resolves-to** between **ipv4-addr** and **mac-addr**
 - belongs-to** between **ipv6-addr** and **autonomous-system**
 - resolves-to** between **ipv4-addr** and **mac-addr**
 - belongs-to** between **ipv6-addr** and **autonomous-system**
- Fixed typos in Extension Definition Additional Examples

- typos were fixed in titles for *section C.2.2 - Adding properties to an existing STIX object instance* and *section C.2.3 - Adding properties to an existing STIX relationship object instance*.
- Special characters were fixed in some participants names in *Appendix F: Acknowledgments*.
- All SCO ids were updated in examples to agree with the `generate_id` method in *python-stix2* library.
- Included all changes based on ITU recommendations.
- Improved references through the document.
 - missing references to sections were added at different places.
 - some references were fixed to point to the right section.
 - in the description of STIX object properties whose value is either a vocabulary or an enumeration, a reference pointing to the given vocabulary or enumeration was added.

2. Conformance

The conformance requirements stated in the OASIS Standard STIX Version 2.1 [STIX-v2.1] are not changed in any way by the publication of this Errata document.

Appendix A: Normative References

The following documents are referenced in such a way that some or all of their content constitutes requirements of this document.

[STIX-v2.1]

STIX Version 2.1. Edited by Bret Jordan, Rich Piazza, and Trey Darley. 10 June 2021. OASIS Standard. <https://docs.oasis-open.org/cti/stix/v2.1/os/stix-v2.1-os.html> (<https://docs.oasis-open.org/cti/stix/v2.1/os/stix-v2.1-os.html>). Latest stage: <https://docs.oasis-open.org/cti/stix/v2.1/stix-v2.1.html> (<https://docs.oasis-open.org/cti/stix/v2.1/stix-v2.1.html>).

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